Sreeram Radhakrishnan

Indian Institute of Technology, Madras

Education

Dual Degree (B.Tech + M.Tech) Electrical Engineering	*Indian Institute of Technology, Madras*	2023 – Present	CGPA 8.68/10
AISSCE (Class XII), CBSE	*Chettinad Vidyashram, Chennai*	2020 – 2022	97.2%
AISSE (Class X), CBSE	*Atomic Energy Central School, Anupuram*	2008 – 2020	96.2%

Publications

- Co-authored, "APEX: A Dual-Sparsity Accelerator for Precise and Efficient SNN Inference," **ISVLSI 2026** [arXiv]. *(Other contributions under preparation for submission to IEEE TVLSI.)*

Scholastic Achievements

- **Gold Medalist** in the **VLSI** Problem Statement released by **ISRO**[1] at the 14th Inter IIT Tech Meet.
- Secured **All India Rank 890** in JEE Advanced 2023 and **All India Rank 747** in JEE (Main) 2023, out of over 1.2 million students nationwide.
- Secured a **100 percentile** in Physics in JEE (Main) 2023 out of over 1.2 million students nationwide.

Internship Experience

Design of a Unified Multi-Precision Transcendental Function Unit for GPUs **May 2026 – Jul 2026**

Skills: SystemVerilog, Synopsys DC *Agrani Labs, A stealth GPU startup*

- Designed the **RTL** of a unified **multi-precision transcendental function unit** for GPUs that computes multiple functions across floating-point formats with shared hardware, and took it through microarchitecture, **synthesis**, **post-synthesis netlist verification** and **formal datapath verification (Synopsys HECTOR)** to deliver a verified block optimized for PPA.

Relevant Coursework and Skills

Coursework : Embedded Memory Design, Computer Organization, Solid State Devices, Digital IC Design, CAD for VLSI, Secure Processor Microarchitecture, Machine Learning Foundations

Skills : SystemVerilog, Cadence Xcelium, Genus, Innovus, Calibre, Virtuoso, Tempus, LTSpice, Vivado HLS, AVR and ARM Assembly, C, C++, Python, Bash

Projects and Experiences

Design of a Digital Dual-Sparse Neuromorphic Accelerator **Jan 2026 – Present**

Skills: SystemVerilog, Synopsys DC, Python *Brainseek Lab, IIT Madras*

- Working on the architectural and microarchitectural design of a **digital dual-sparse neuromorphic accelerator** being built end-to-end with its accompanying software stack, under the guidance of **Prof. Gopalakrishnan Srinivasan** at Brainseek Lab, IIT Madras.
- Designed a custom **dimension-agnostic instruction set architecture (ISA)** from scratch, tailored to the accelerator's sparsity and precision requirements; mapped quantized SNN model weights and parameters to a unified, **fully integer-arithmetic mixed-precision datapath**, building the corresponding hardware to support quantized inference.
- Built a **hardware implementation of the PASC-IF neuron** as a fully combinational datapath with no added latency, letting the converted SNN reach ANN-equivalent accuracy at far fewer timesteps and substantially lower energy. Published as **APEX** at **ISVLSI 2026**; further work under preparation for IEEE TVLSI.
- Ran a systematic **dataflow design-space exploration** for the accelerator's PE array, deriving a closed-form model of on-chip SRAM traffic, off-chip traffic, and storage overhead tradeoffs that unifies multiple candidate dataflows; identified an optimal schedule and array shape that substantially reduces SRAM reads relative to the baseline design at lower per-PE storage cost, validated across VGG-16, ResNet-18, and AlexNet.

[1] Indian Space Research Organisation

Inter IIT Tech Meet – ASIC Physical Design **Nov 2025 – Dec 2025**

Skills: Verilog, Xcelium, Genus, Innovus, Calibre, Virtuoso, Tempus *IIT Madras*

- Implemented the **RTL-to-GDSII ASIC flow** using Cadence tools for synthesis, place-and-route, and post-layout simulations, achieving a **DRC- and LVS-clean** layout for the **decimation filter of a ΔΣ ADC** in **SCL180.**
- Performed **post-layout STA signoff** in Cadence **Tempus** across SS/FF PVT corners with single-mode and OCV analysis, achieving positive setup and hold slack across all modes.
- Optimized area and power through architectural and physical-design choices, achieving improved PPA with validated post-layout performance using back-annotated simulations.

Digital IC Design Lab **Jul 2025 – Nov 2025**

Skills learnt: SystemVerilog, NGSpice, C, Python *IIT Madras*

- Designed and simulated CMOS logic circuits (16-bit ripple carry adder, ring oscillator) in **NGSpice** with sky130 PDK.
- Created physical layouts in **KLayout**, performed **DRC** and **LVS** checks, and validated schematic-to-layout consistency.
- Implementation of **carry-look ahead adder architecture (Brent-Kung, Kogge-stone)** for delay and frequency.
- Implementation of **a 16-bit carry-bypass adder in SystemVerilog and STA** to identify critical path and slack through constraint-based timing tools.

Enabling Use-Case Adaptation in Speech Recognition Systems **Sep 2024 – Sep 2025**

Skills learnt: Python, Pytorch, Pandas, Bash *AI4Bharat Lab, IIT Madras*

- Working under the guidance of **Prof. Mitesh Khapra** on developing an agentic framework for synthetic text generation to improve the adaptability of **Automatic Speech Recognition (ASR) systems** to specialized domains and **low-resource language** scenarios.
- Comparing a novel synthetic text generation technique against text only methods like Shallow fusion with external LMs, hotword boosting, and LLM rescoring to assess ASR performance improvements.

Project NightVision **May 2024 – Dec 2024**

Skills learnt: Deep Learning, PyTorch, Numpy, Pandas *IIT Madras*

- Worked on Low light image enhancement leveraging Computer Vision and various Deep Learning methods.
- Performed an extensive literature review on Retinex theory, Attention mechanism and various other methods.
- Implemented fundamental architectures like Vision Transformer, U-Net and GANs from scratch using PyTorch.
- Implemented a Retinex-based deep learning model featuring iterative illumination refinement and reflectance denoising for low-light image enhancement.

Leadership / Extracurricular Activities

AI Club Coordinator **May 2024 – Mar 2025**

Skills learnt: Deep Learning, PyTorch, Numpy, Pandas *IIT Madras*

- Conducted sessions on basics of Machine Learning and Computer Vision to 1000+ students from all over India.
- Conducted workshop at various colleges on Deep Learning and Advanced Vision architectures.
- Spearheaded a project on Image Captioning models exploring various architectures and implementing them.

Avanti Mentor **Nov 2023 – Oct 2024**

Avanti Fellows *IIT Madras Chapter*

- Mentored students from Jawahar Navodaya Vidyalaya, Puducherry and Barwani for JEE Mains and Advanced.
- Prepared mock examination papers to meet the specific requirements of diverse competitive examinations.

IViL Volunteer **Dec 2024 – Jan 2025**

IIT for Villages *Shaastra & Saarang, IIT Madras*

- Empowered rural women with business skills by managing the logistics and finances at the IViL (IIT for Villages) food stalls during the Shaastra and Saarang fests at IIT Madras.